Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:

- Load the packages `tidyverse`, `rvest`, and any others you end up using.
- Check your working directory

```
#1

library(tidyverse)
library(lubridate)
library(here); here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
library(rvest)
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham's 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to https://www.ncwater.org/WUDC/app/LWSP/search.php
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```r
#2
webpage <- read_html('https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024')
webpage
```

```
## {html_document}
## <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
## [1] <head>\n<title>DWR :: Local Water Supply Planning</title>\n<meta http-equ ...
## [2] <body id="plan">\r\n<!--<div id="division-header">\r\n<a name="top" href= ...
```

3. The data we want to collect are listed below:

- From the "1. System Information" section:
- Water system name
- PWSID
- Ownership
- From the "3. Water Supply Sources" section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

> HINT: The first value should be "Durham", the second "03-32-010", the third "Municipality", and the last should be a vector of 12 numeric values (represented as strings)".

```r
#3
water_system_name <- webpage %>%
  html_nodes('div+ table tr:nth-child(1) td:nth-child(2)') %>%
  html_text()
water_system_name
```

```
## [1] "Durham"
```

```r
PWSID <- webpage %>%
  html_nodes('td tr:nth-child(1) td:nth-child(5)') %>%
  html_text()
PWSID
```

```
## [1] "03-32-010"
```

```r
Ownership <- webpage %>%
  html_nodes('div+ table tr:nth-child(2) td:nth-child(4)') %>%
  html_text()
Ownership
```

```
## [1] "Municipality"
```

```r
MGD <- webpage %>%
  html_nodes('th~ td+ td') %>%
  html_text()%>%
  as.numeric()
MGD
```

```
##  [1] 34.50 36.06 37.33 32.10 46.65 37.36 38.20 41.90 36.58 36.73 42.96 34.45
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

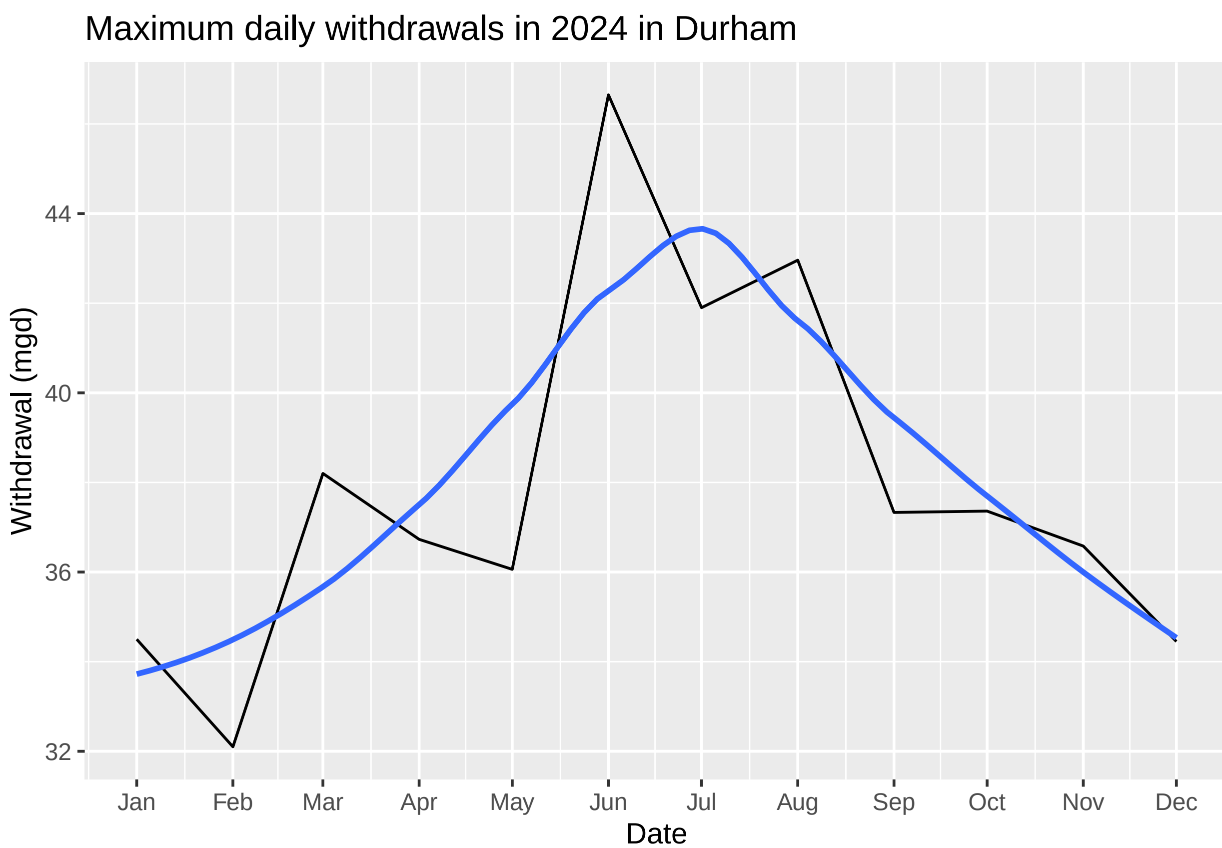
   TIP: Use `rep()` to repeat a value when creating a dataframe.

   NOTE: It's likely you won't be able to scrape the monthly widthrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc…

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```r
#4

#Create a dataframe of withdrawals
df_withdrawals <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
                             "Year" = rep(2024,12))

df_withdrawals <- df_withdrawals %>%
  mutate(water_system_name= !!water_system_name,
         PWSID = !!PWSID,
         Ownership = !!Ownership,
         MGD = !!MGD,
         Date = my(paste(Month,"-",Year)))
#5
#Plot
ggplot(df_withdrawals,aes(x=Date,y=MGD)) +
  geom_line() +
  geom_smooth(method="loess",se=FALSE) +
  scale_x_date(date_breaks = "1 month", date_labels = "%b")+
  labs(title = paste("Maximum daily withdrawals in 2024 in",water_system_name),
       y="Withdrawal (mgd)",
       x="Date")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - "PWSID" and "year" - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function's output

```
#6.

scrape.it <- function(PWSID, Year){

  #Retrieve the website contents
  the_website <- read_html(paste0('https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid='
                                  ,PWSID,'&year=', Year))

  #Set the element address variables
  water_system_name_tag <- 'div+ table tr:nth-child(1) td:nth-child(2)'
  PWSID_tag <- 'td tr:nth-child(1) td:nth-child(5)'
  Ownership_tag <- 'div+ table tr:nth-child(2) td:nth-child(4)'
  MGD_tag <- 'th~ td+ td'

  #Scrape the data items
  water_system_name <- the_website %>% html_nodes(water_system_name_tag) %>% html_text()
```

```
  PWSID<- the_website %>%   html_nodes(PWSID_tag) %>%  html_text()
  Ownership <- the_website %>% html_nodes(Ownership_tag) %>% html_text()
  MGD <- the_website %>% html_nodes(MGD_tag) %>% html_text()

  #Convert to a dataframe
  df_withdrawals_2 <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
                                 "Year" = rep(Year,12),
                                 "Max Day Use" = as.numeric(MGD)) %>%
    mutate(water_system_name = !!water_system_name,
           PWSID = !!PWSID,
           Ownership = !!Ownership,
           Date = my(paste(Month,"-",Year)))

  #Return the dataframe
  return(df_withdrawals_2)
}
```

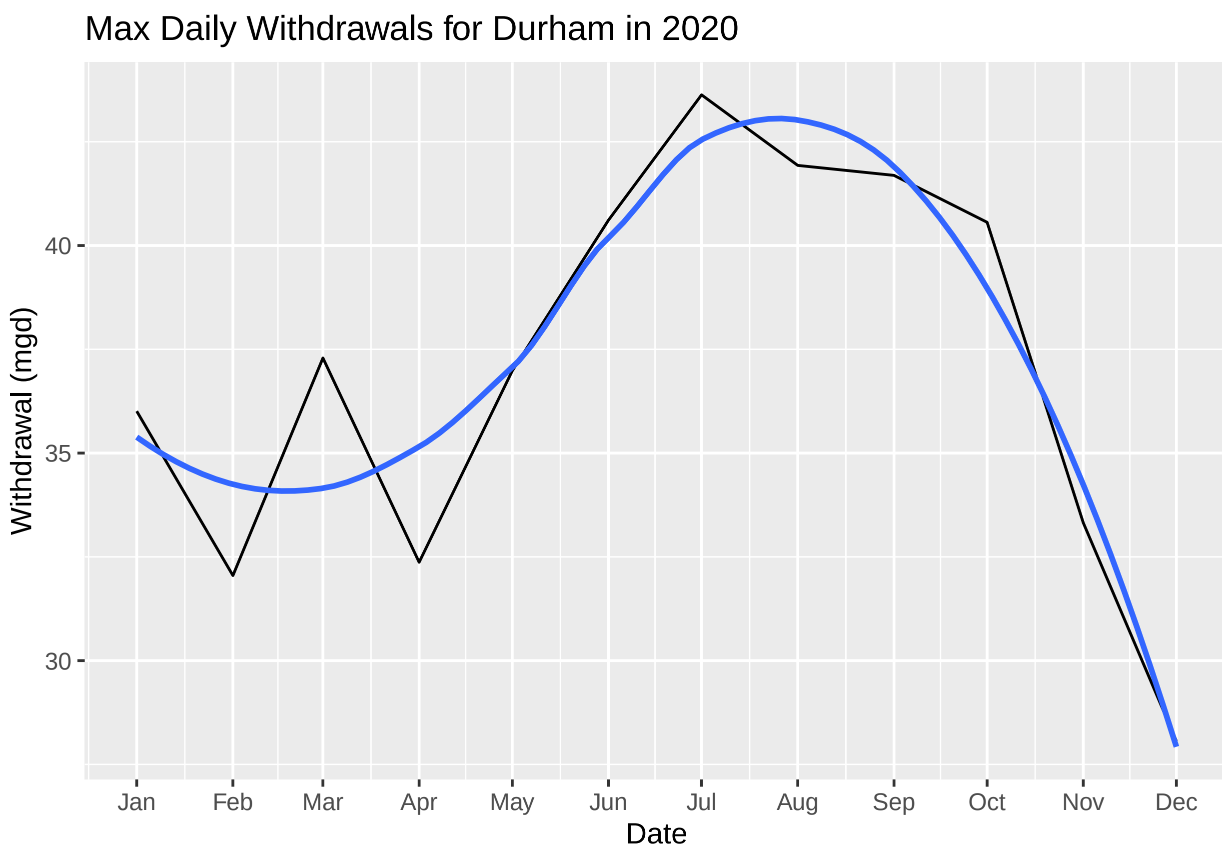
7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```
#7
Year = 2020
PWSID= '03-32-010'

the_dfs <- map(Year,scrape.it,PWSID=PWSID)
df_durham <- bind_rows(the_dfs)
str(df_durham)
```

```
## 'data.frame':    12 obs. of  7 variables:
##  $ Month            : num  1 5 9 2 6 10 3 7 11 4 ...
##  $ Year             : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
##  $ Max.Day.Use      : num  36 37 41.7 32 40.6 ...
##  $ water_system_name: chr  "Durham" "Durham" "Durham" "Durham" ...
##  $ PWSID            : chr  "03-32-010" "03-32-010" "03-32-010" "03-32-010" ...
##  $ Ownership        : chr  "Municipality" "Municipality" "Municipality" "Municipality" ...
##  $ Date             : Date, format: "2020-01-01" "2020-05-01" ...
```

```
ggplot(df_durham,aes(x=Date,y=`Max.Day.Use`)) +
  geom_line() +
  geom_smooth(method="loess",se=FALSE) +
  scale_x_date(date_breaks = "1 month", date_labels = "%b")+
  labs(
    title = paste(
      "Max Daily Withdrawals for", water_system_name,"in",Year),
    y="Withdrawal (mgd)",
    x="Date"
    )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

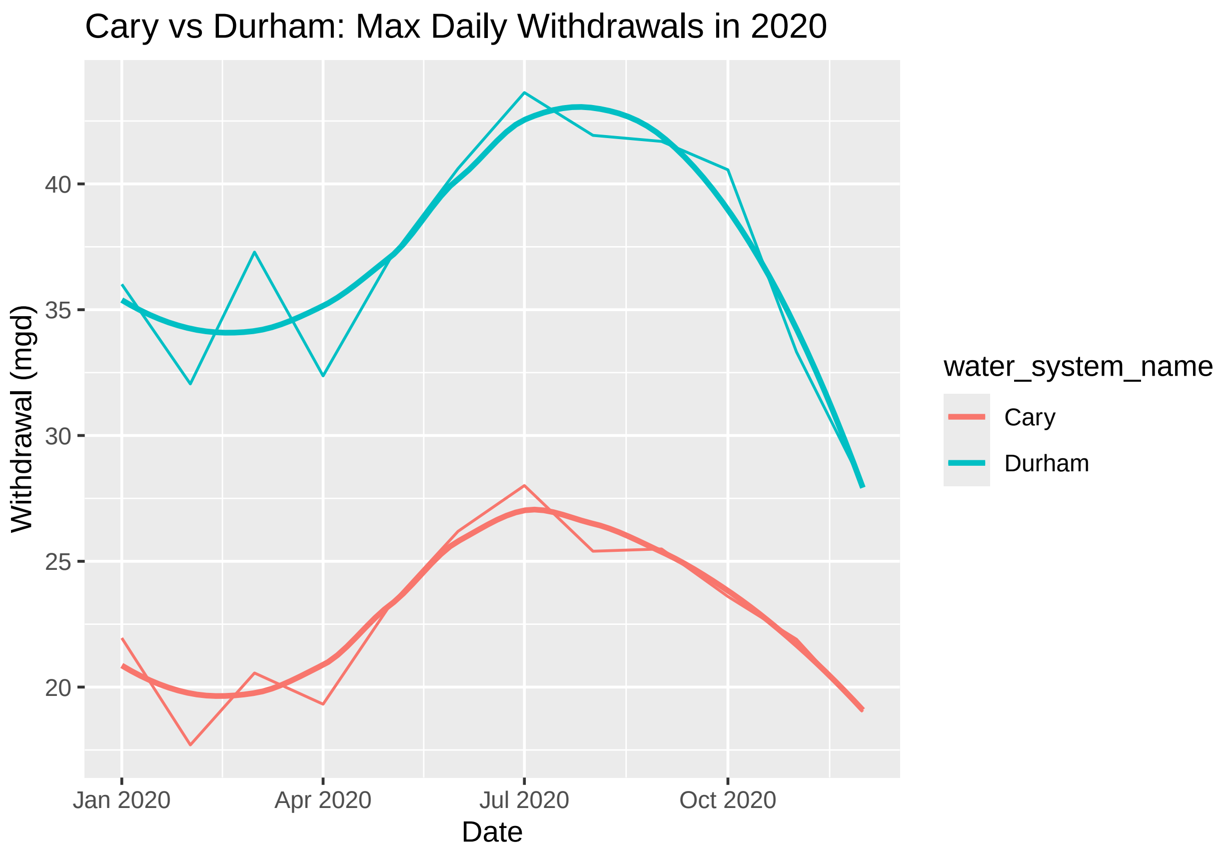
```
#8
#Extract for Cary
Year = 2020
PWSID= '03-92-020'

cary <- map(Year,scrape.it,PWSID=PWSID)
df_cary <- bind_rows(cary)
str(df_cary)
```

```
## 'data.frame':    12 obs. of  7 variables:
##  $ Month            : num  1 5 9 2 6 10 3 7 11 4 ...
##  $ Year             : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
##  $ Max.Day.Use      : num  21.9 23.2 25.5 17.7 26.2 ...
##  $ water_system_name: chr  "Cary" "Cary" "Cary" "Cary" ...
##  $ PWSID            : chr  "03-92-020" "03-92-020" "03-92-020" "03-92-020" ...
##  $ Ownership        : chr  "Municipality" "Municipality" "Municipality" "Municipality" ...
##  $ Date             : Date, format: "2020-01-01" "2020-05-01" ...
```

```
#combine df
df_cary_durham  <- bind_rows(df_cary,df_durham)
```

```r
#Plot
ggplot(df_cary_durham,aes(x=Date,y=`Max.Day.Use`,color=water_system_name)) +
  geom_line() +
  geom_smooth(method="loess",se=FALSE) +
  labs(title =paste("Cary vs Durham: Max Daily Withdrawals in",Year),
       y="Withdrawal (mgd)",
       x="Date")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023.Add a smoothed line to the plot (method = 'loess').

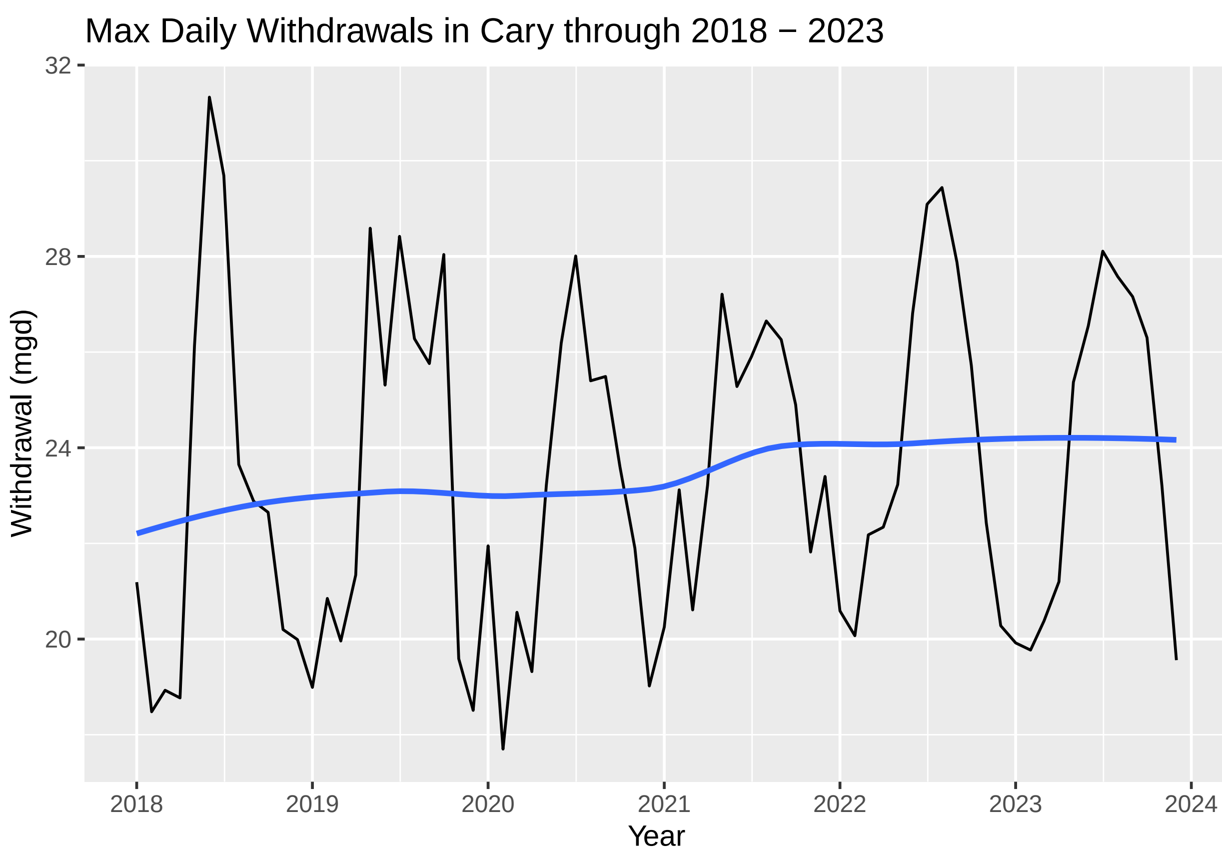
   TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to `bind_rows()` to combine the dataframes into a single one, and use that to construct your plot.

```r
#9
#Extract for Cary
Year = c(2018:2023)
PWSID= '03-92-020'

cary <- map2(PWSID, Year, scrape.it) %>%
  bind_rows()
str(cary)
```

```
## 'data.frame':    72 obs. of  7 variables:
##  $ Month            : num  1 5 9 2 6 10 3 7 11 4 ...
##  $ Year             : int  2018 2018 2018 2018 2018 2018 2018 2018 2018 2018 ...
##  $ Max.Day.Use      : num  21.2 26.1 22.9 18.5 31.3 ...
##  $ water_system_name: chr  "Cary" "Cary" "Cary" "Cary" ...
##  $ PWSID            : chr  "03-92-020" "03-92-020" "03-92-020" "03-92-020" ...
##  $ Ownership        : chr  "Municipality" "Municipality" "Municipality" "Municipality" ...
##  $ Date             : Date, format: "2018-01-01" "2018-05-01" ...
```

```r
#Plot
ggplot(cary,aes(x=Date,y=`Max.Day.Use`)) +
  geom_line() +
  geom_smooth(method="loess",se=FALSE) +
  scale_x_date(date_breaks = "1 year",date_labels = "%Y")+
  labs(title = paste("Max Daily Withdrawals in Cary through",
       Year[1],"-",Year[length(Year)]),
       y="Withdrawal (mgd)",
       x="Year")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: It seems that Cary do have a strong seasonality in water usage over time. Cary's water withdrawals surge in summer season while there is no distinctive long term trend. >